

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour
Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Code of Conduct:

*I S.Mahammed Nayeem (192425294)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

*S.Mahammed Nayeem
Signature of the student:*

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided

Question 1 : Error Analysis Task

Given:

- Training Mean Squared Error (MSE) = **0.85**
- Testing Mean Squared Error (MSE) = **5.20**

Analysis:

Mean Squared Error (MSE) is a commonly used performance metric for regression models. It measures the average squared difference between the actual and predicted values. A lower MSE indicates better prediction accuracy.

In this case, the training MSE is **0.85**, while the testing MSE is **5.20**. The large difference between these values shows that the neural network performs well on the training dataset but poorly on unseen testing data. This indicates that the model has learned the training data too closely and is unable to generalize effectively. Such a condition is called **overfitting**.

Possible Reasons:

- The neural network has memorized the training data instead of learning general patterns.
- The training dataset is too small or does not represent all possible data variations.
- The model contains too many hidden layers or neurons, making it excessively complex.
- Regularization techniques such as Dropout or L2 Regularization were not used.
- The model was trained for too many epochs, causing it to overfit.
- The training and testing datasets may have different distributions.
- Input data may not have been properly normalized or preprocessed.

Effects of Overfitting:

- High accuracy on training data.
- Poor prediction accuracy on new or unseen data.
- Increased testing error.
- Reduced reliability of the model in real-world applications.
- Poor generalization capability.

Corrective Measures:

- Increase the size of the training dataset.
- Apply Dropout layers to reduce overfitting.
- Use L1 or L2 Regularization.
- Implement Early Stopping to stop training before overfitting occurs.
- Reduce the number of hidden layers or neurons.
- Perform feature selection to remove unnecessary features.
- Normalize or standardize the input data.
- Tune hyperparameters such as learning rate, batch size, and number of epochs.
- Use cross-validation to evaluate model performance.

Conclusion:

The large difference between the training MSE (**0.85**) and testing MSE (**5.20**) indicates that the neural network is **overfitting**. Although the model performs well on the training data, it does not generalize well to unseen data. By increasing the training data, simplifying the model, applying regularization techniques, using early stopping, and tuning hyperparameters, the testing error can be reduced and the overall performance of the neural network can be improved.

Python Code :

```
training_mse = 0.85
testing_mse = 5.20

print("Training MSE:", training_mse)
print("Testing MSE:", testing_mse)

difference = testing_mse - training_mse
print("Difference:", difference)

if testing_mse > training_mse:
    print("\nAnalysis")
    print("The neural network is overfitting.")

    print("\nPossible Reasons")
    print("1. Overfitting")
    print("2. Small training dataset")
    print("3. High model complexity")
    print("4. Lack of regularization")
    print("5. Excessive training epochs")
    print("6. Different training and testing data distribution")

    print("\nCorrective Measures")
    print("1. Increase training data")
    print("2. Apply Dropout")
    print("3. Use L2 Regularization")
    print("4. Apply Early Stopping")
    print("5. Reduce model complexity")
    print("6. Tune hyperparameters")
    print("7. Normalize input data")
    print("8. Use cross-validation")
else:
    print("The model generalizes well.")
```

Sample Output :

```
Training MSE: 0.85
Testing MSE: 5.2
Difference: 4.35
```

Analysis :

The neural network is overfitting.

Possible Reasons:

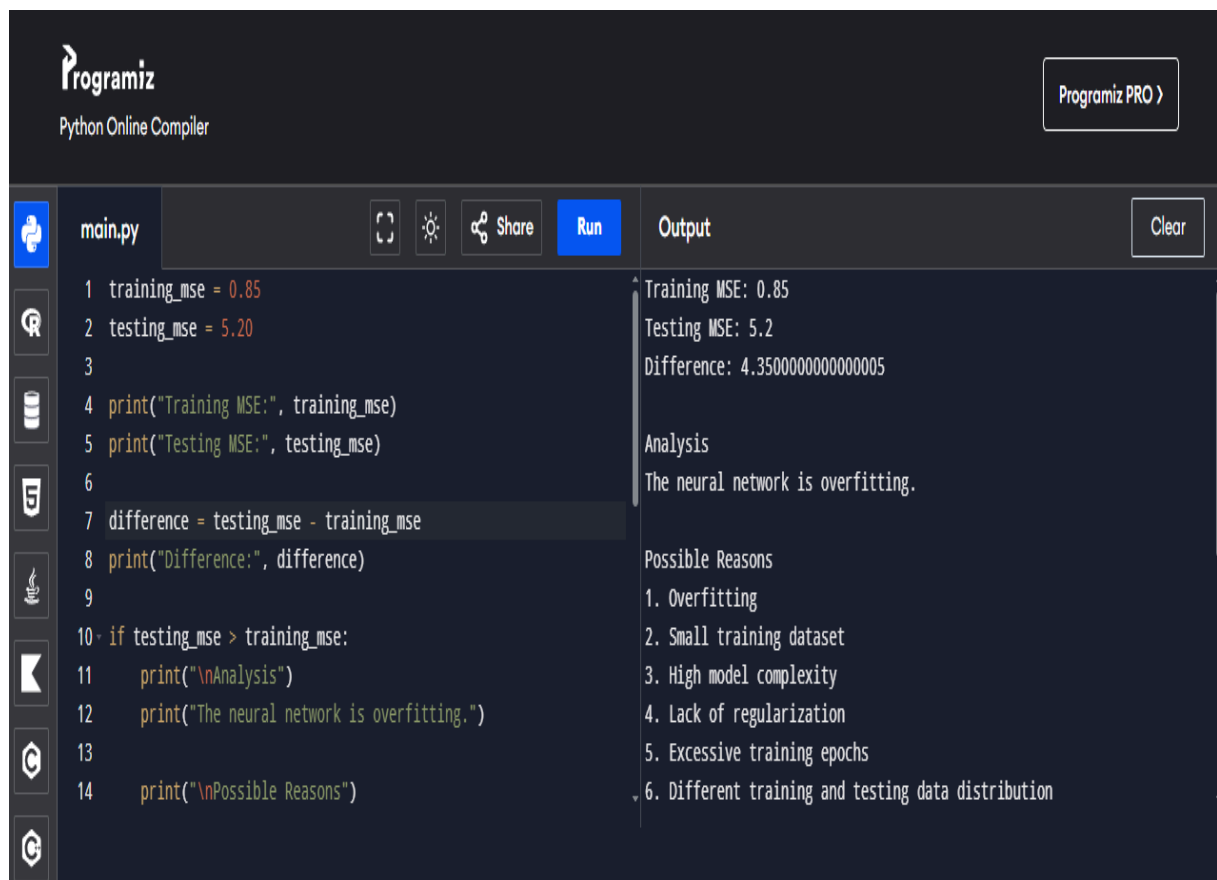
1. Overfitting
2. Small training dataset
3. High model complexity
4. Lack of regularization
5. Excessive training epochs
6. Different training and testing data distribution

Corrective Measures:

1. Increase training data
2. Apply Dropout
3. Use L2 Regularization
4. Apply Early Stopping
5. Reduce model complexity
6. Tune hyperparameters
7. Normalize input data
8. Use cross-validation

Result:

1. The training MSE is **0.85**, while the testing MSE is **5.20**.
2. The higher testing MSE indicates that the model is **overfitting**.
3. The model performs well on training data but poorly on unseen data.



The screenshot shows the Programiz Python Online Compiler interface. The code editor on the left contains a Python script named `main.py` with the following lines:

```
1 training_mse = 0.85
2 testing_mse = 5.20
3
4 print("Training MSE:", training_mse)
5 print("Testing MSE:", testing_mse)
6
7 difference = testing_mse - training_mse
8 print("Difference:", difference)
9
10 if testing_mse > training_mse:
11     print("\nAnalysis")
12     print("The neural network is overfitting.")
13
14 print("\nPossible Reasons")
```

The output panel on the right displays the results of the script execution:

```
Training MSE: 0.85
Testing MSE: 5.2
Difference: 4.3500000000000005

Analysis
The neural network is overfitting.

Possible Reasons
1. Overfitting
2. Small training dataset
3. High model complexity
4. Lack of regularization
5. Excessive training epochs
6. Different training and testing data distribution
```